CONF42 · PROMPT ENGINEERING 2026

Teaching machines to spot bias in the news

From a five-LLM pipeline to our own bias detection model

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WHY THIS MATTERS

Every headline carries a point of view

Word choice

“Protesters” or “rioters”? One word changes the story.

Framing

What leads the article, and whose voice is quoted first.

Omission

What’s left out can matter as much as what’s said.

Our goal: detect linguistic bias automatically, at news scale, and explain it.

THE JOURNEY

Six stages, in the order we built them

01

Collect

X trends + Google News

02

Deduplicate

One story, one record

03

Ask 5 LLMs

One wrapper, five models

04

Expert review

PhD linguists in the loop

05

Train our own

500K stories, 70 biases

06

Keep improving

Monthly human review

01 · COLLECT

Two sources: what people talk about, and what gets reported

X (Twitter) API

- Trending topics
- Latest breaking news
- Tells us what to look for

Google News APIs

- Full articles on each trend
- Many outlets, one event
- Gives us depth and contrast

02 · DEDUPLICATE

One story, many headlines

Duplicates

The same event reported by many outlets, or the same article syndicated and re-posted.

➜ **Grouped into one event**, keeping each outlet's version for comparison.

Continuations

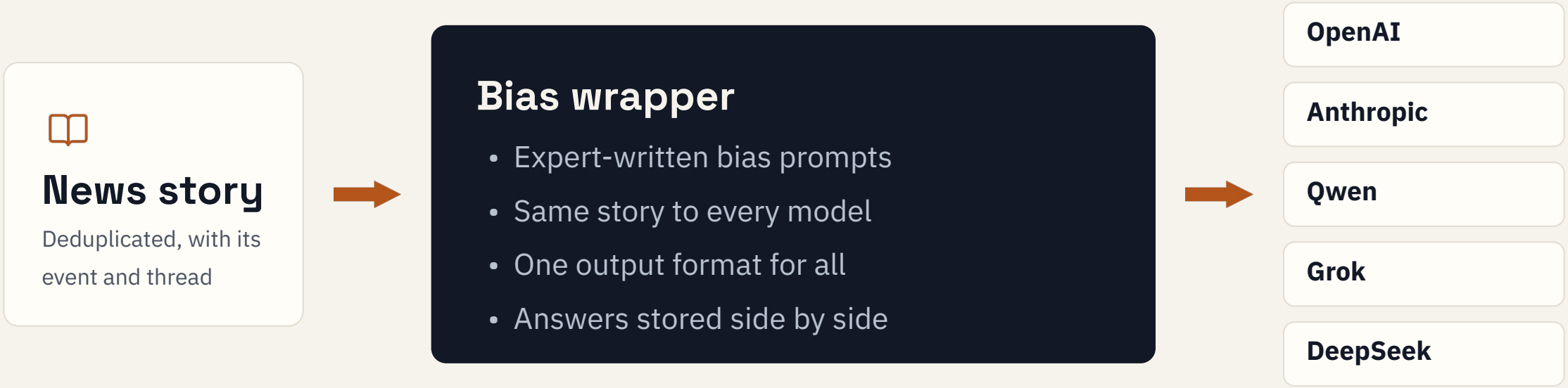
The same story evolving over days: breaking news, then updates, then follow-ups.

➜ **Linked into one thread**, so we see how the framing shifts over time.

Without this, the dataset over-counts big stories and the model learns the news cycle, not bias.

03 · ASK THE LLMS

A wrapper: one question, five models



03 · WHAT WE LEARNED

Same story, five readers, different verdicts

Agreement

When all five flag the same bias, we can be fairly confident.

Disagreement

When they split, the text is genuinely ambiguous — or a model has a blind spot.

Inconsistency

Even one model can answer differently on a second run.

Disagreement isn’t noise. It tells you where humans need to look.

04 · HUMANS IN THE LOOP

PhD linguists defined what “bias” means

Define

A taxonomy of linguistic biases, and the prompts that ask for them

Collect

Choose stories that cover every bias, topic and outlet fairly

Annotate

Label spans, settle model disagreements, set the gold standard

Specialists in **bias and hate speech**, not general annotators.

04 · SPECIALIST MODELS

Small models, each an expert in one bias

Open datasets

Public, labelled data for specific bias and hate-speech types

Specialists

One small classifier per kind of bias

Second opinion

Check every LLM label before it enters the dataset

LLM labels + specialist checks + expert review = data we could trust.

05 · THE DATASET

500K+
balanced stories of our own

70
linguistic biases labelled

5
major LLMs per story, checked by our models

Enough to train a bias detection model of our own.

05 · TRAIN OUR OWN

The LLMs became teachers, not the product

It learned from

- Labels where the 5 LLMs agreed
- Expert rulings where they didn’t
- Checks from specialist models

It does

- Reads each sentence in context
- Checks all 70 biases in one pass
- Runs on our own infrastructure

05 · WHAT IT RETURNS

Not a yes/no label: a span, a severity and a reason

ILLUSTRATIVE SENTENCE

“The minister finally admitted the plan had failed.”

BIAS
Loaded verb

SEVERITY
Graded, e.g. moderate

REASON
Implies guilt and delay

NEUTRAL REWRITE
“The minister said the plan had not worked.”

05 · BENCHMARK

Our model vs. the five-LLM pipeline

Measure	5-LLM pipeline	Our model
Consistency	Varies by model and by run	Same input, same answer
Speed	5 API calls per story	One pass · 5s per story
Cost	5× paid tokens per story	3% of pipeline cost
Agreement	Baseline	with expert labels

More consistent. Faster. Cheaper.

06 · KEEP IMPROVING

Every month, the loop runs again

1
Sample
Take a fresh sample of this month's output

→

2
Review
Experts analyse it by hand

→

3
Diagnose
Find errors, anomalies and new patterns

→

4
Retrain
Update the dataset, retrain, re-benchmark

Language changes. The model has to keep up.

TAKEAWAYS

Four lessons for prompt engineers

Experts write the best prompts

Domain knowledge beats prompt tricks.

Ask more than one model

Disagreement shows you where to look.

Use LLMs as teachers

Label with big models, run a small one.

Ship the loop, not the model

Human review every month keeps it honest.

THANK YOU

Questions?

Collect → deduplicate → ask five LLMs → expert review → train our own → retrain monthly

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